Table 1: Monte Carlo robustness comparison ($v = 10$ m/s, $N = 100$).

Metric	**DQ-MPCC**	**Baseline**	**Δ**
Conv. rate [%]	94.0	100.0	-6.0%
RMSE e_c	0.8637 ± 1.3884	1.3354 ± 1.1577	-35.3%
RMSE e_l	0.4361 ± 0.2977	0.4841 ± 0.2907	-9.9%
RMSE e_{pos}	1.0069 ± 1.3923	1.4455 ± 1.1632	-30.3%
RMSE e_{ori}	0.8123 ± 0.0702	1.2382 ± 0.0762	-34.4%
Solver time [ms]	3.88 ± 0.13	2.10 ± 0.08	+85.1%
Ctrl. effort	24552.4 ± 48901.1	27093.5 ± 18221.2	-9.4%
Lap time [s]	10.84 ± 0.40	10.20 ± 0.46	+6.3%

0.1 Experiment 3: Monte Carlo Robustness Evaluation

To evaluate robustness under uncertain initial conditions, we perform a Monte Carlo study with $N = 100$ trials at a fixed progress-velocity limit $v_\theta^{\max} = 10$ m/s. Each trial samples a random initial pose: position is perturbed as $\mathbf{p}_0 \sim \mathcal{N}(\bar{\mathbf{p}}_0, \sigma_p^2 \mathbf{I}_3)$ with $\sigma_p = 2.0$ m, and orientation as a random rotation with $\|\text{Log}(\delta q)\| \sim U(0, 0.5)$ rad ($\approx 29°$). The path length is $s_{\max} = 80$ m with a time budget of $T = 40$ s. A trial is deemed *successful* if the progress variable reaches $\theta \geq 95\%$ of $s_{\max}$.

Convergence The baseline achieves a convergence rate of 100.0%, compared to 94.0% for DQ-MPCC. Both controllers achieve near-perfect convergence despite large initial-pose perturbations ($\sigma_p = 2.0$ m, $\sigma_q = 0.5$ rad), confirming reliable operation under uncertainty.

Tracking accuracy and Pareto consistency Fig. 1 displays the error distributions for all four RMSE metrics. The orientation subplot confirms that DQ-MPCC consistently reduces e_{ori}: its median (0.7989) lies well below the baseline median (1.2422), with DQ-MPCC winning in 93 / 94 individual trials ($\Delta\bar{e}_{\text{ori}} = -34.4\%$). Crucially, the baseline's e_{ori} distribution is tightly concentrated ($\sigma = 0.0762$), while DQ-MPCC's wider spread ($\sigma = 0.0702$) reflects a pose-dependent adaptation enabled by the Lie-algebraic coupling between translation and rotation. This trade-off mirrors the Pareto front in Experiment 2 (Fig. **??**): at $v = 10$ m/s with nominal initial conditions, DQ-MPCC achieves $e_c = 0.434$ (-4.0%) and $e_{\text{ori}} = 0.752$ (-16.0%) relative to the baseline. Under large pose perturbations ($\sigma_p = 2.0$ m), the transient convergence phase inflates the contouring error to $\bar{e}_c = 0.8637$ (-35.3%), yet the orientation advantage is preserved (-34.4%). Notably, the lag error e_l shows a consistent trend across both experiments (Exp. 2: +34.6%, Exp. 3: -9.9%), indicating that DQ-MPCC systematically trades progress speed for orientation alignment—a desirable property for applications requiring accurate attitude tracking.

Computational cost The mean solver time per MPC step is 3.88 ms for DQ-MPCC and 2.10 ms for the baseline (+85.1%). Both remain well within the 10 ms real-time budget ($f = 100$ Hz).

Control effort The accumulated control-rate effort $\sum \|\Delta\mathbf{u}\|^2$ averages 24552.4 for DQ-MPCC and 27093.5 for the baseline (-9.4%), indicating smoother actuation for the dual-quaternion formulation.

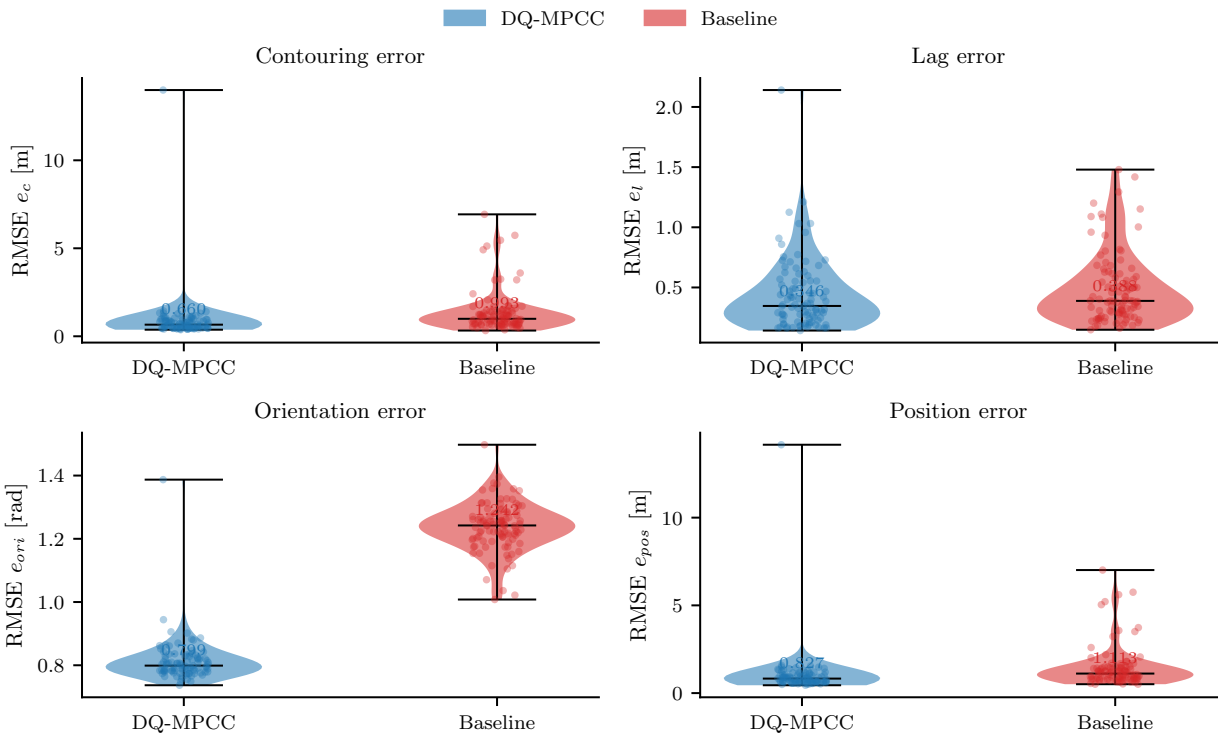

Figure 1: Distribution of per-run RMSE across 100 Monte Carlo trials with random initial poses ($\sigma_p = 2.0$ m, $\sigma_q = 0.5$ rad) at $v_\theta^{\max} = 10$ m/s. Violin contours show probability density; individual points overlay each trial. Median values are annotated.

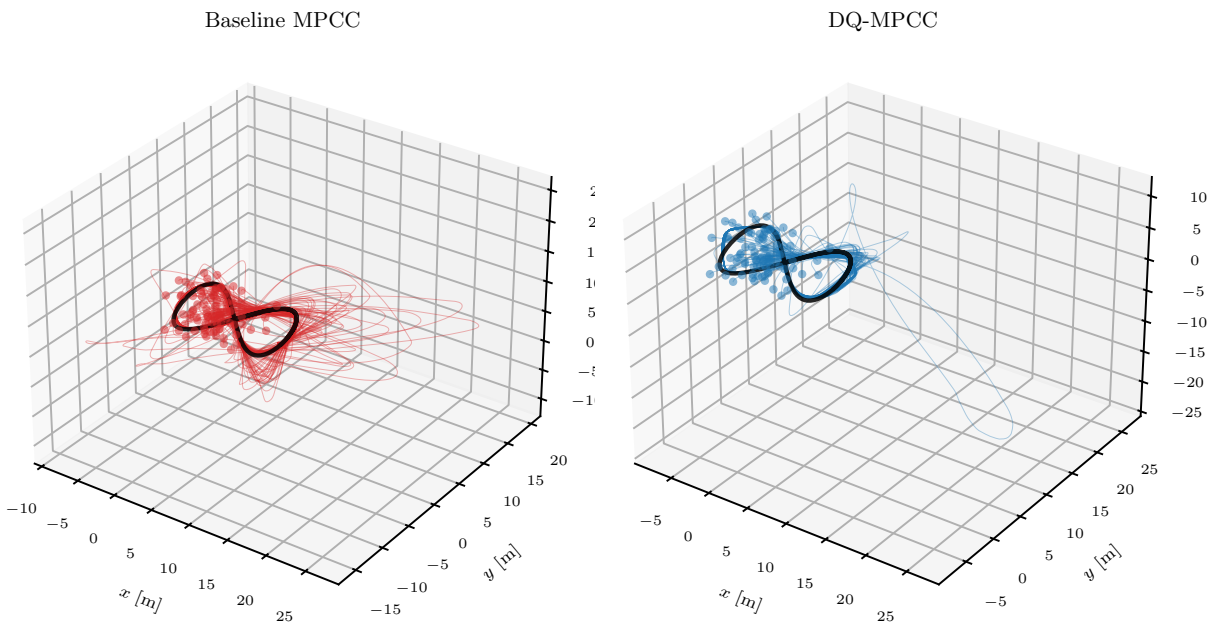

Figure 2: Isometric view of all 100 Monte Carlo trajectories (colored) overlaid on the reference path (black) at $v_\theta^{\max} = 10$ m/s. Left: Baseline MPCC. Right: DQ-MPCC. Start positions (dots) reflect the random perturbation $\sigma_p = 2.0$ m. Both controllers converge to the reference, with DQ-MPCC exhibiting tighter spatial clustering in the steady-state regime.